

PHYS 230 - Electricity and Magnetism

Lecture 2I

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

4 Capacitance and Dielectrics

- Dielectrics (Continued)

5 Current, Resistance, and Electromotive Force

- Electric Current

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Dielectric Breakdown

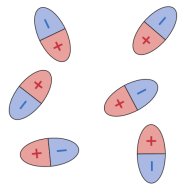
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- **Lightning** is an example of dielectric breakdown in air

Molecular Model of Induced Charge

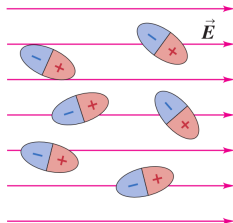
Figure 24.17 Polar molecules (a) without and (b) with an applied electric field $\vec{E}$.

(a)



In the absence of an electric field, polar molecules orient randomly.

(b)

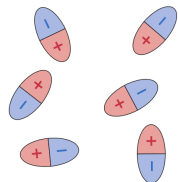


When an electric field is applied, the molecules tend to align with it.

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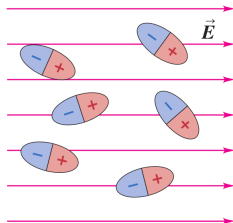
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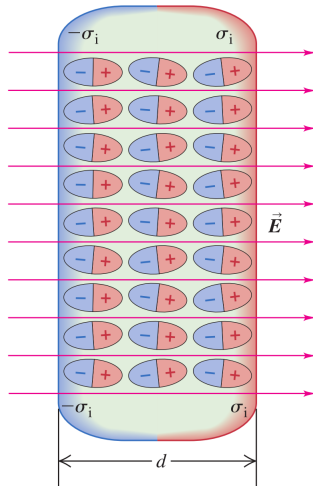
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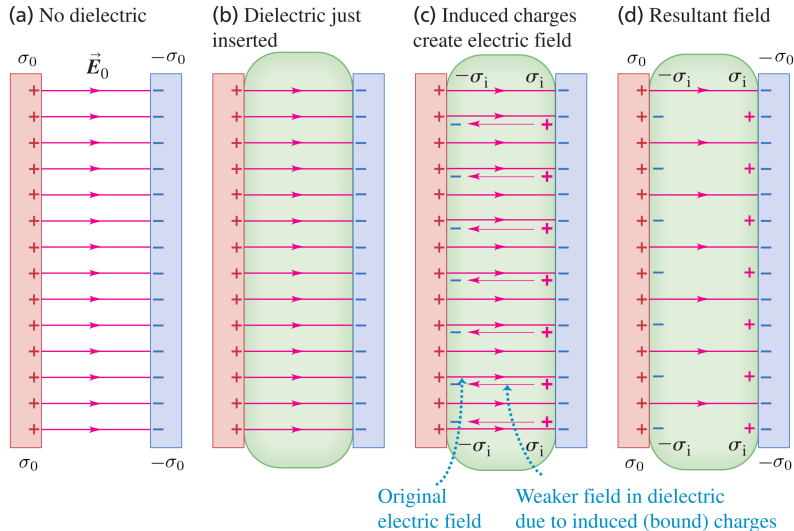
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Figure 24.19 Polarization of a dielectric in an electric field $\vec{E}$ gives rise to thin layers of bound charges on the surfaces, creating surface charge densities σ_i and $-\sigma_i$. The sizes of the molecules are greatly exaggerated for clarity.



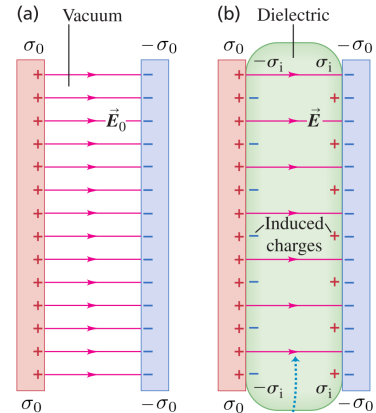
Molecular Model of Induced Charge

Figure 24.20 (a) Electric field of magnitude E_0 between two charged plates. (b) Introduction of a dielectric of dielectric constant K . (c) The induced surface charges and their field. (d) Resultant field of magnitude E_0/K .



Gauss's Law in Dielectrics

Figure 24.15 Electric field lines with
(a) vacuum between the plates and
(b) dielectric between the plates.



For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

Gauss's Law in Dielectrics

Figure 24.22 Gauss's law with a dielectric. This figure shows a close-up of the left-hand capacitor plate in Fig. 24.15b. The Gaussian surface is a rectangular box that lies half in the conductor and half in the dielectric.

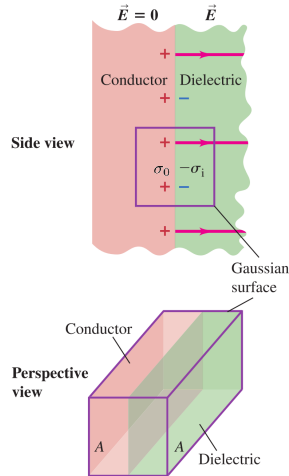
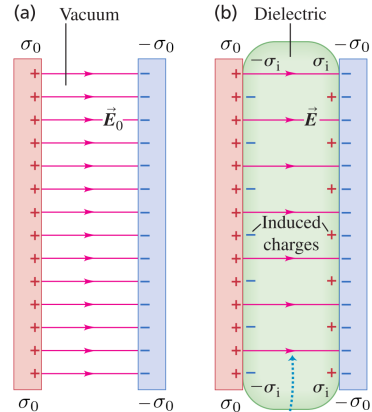


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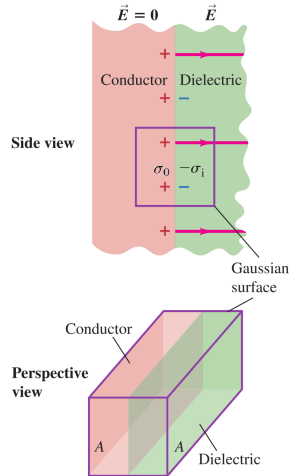
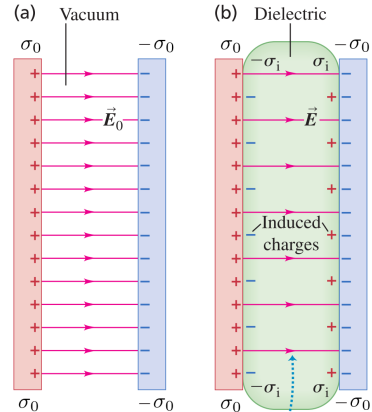


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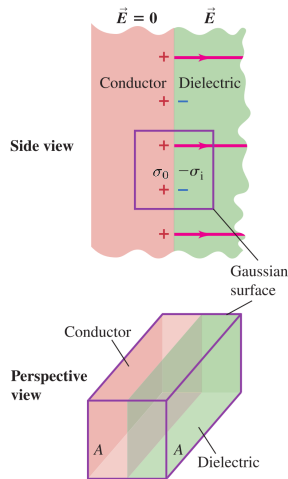
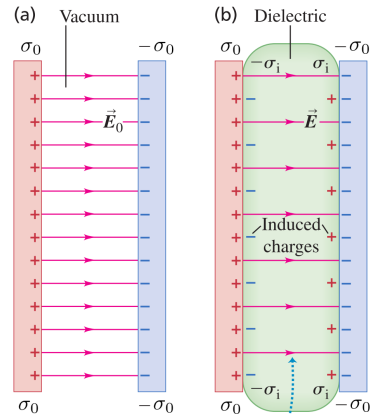


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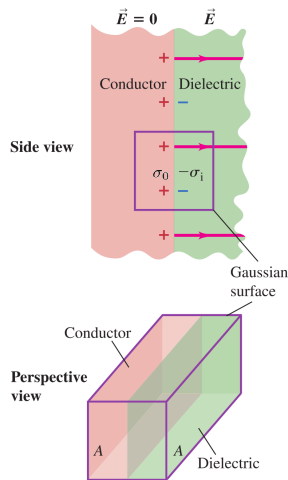
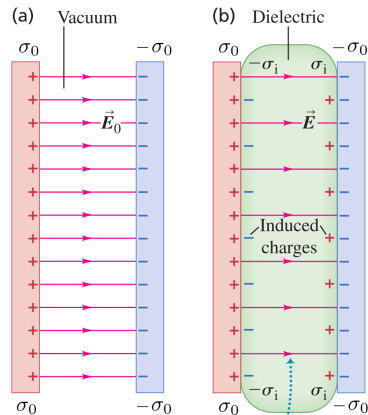


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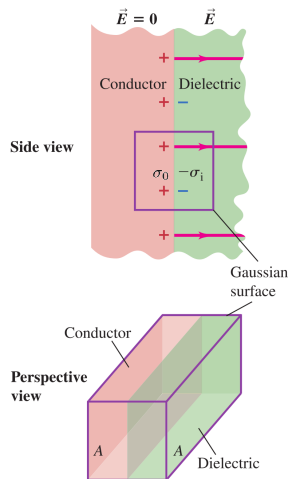
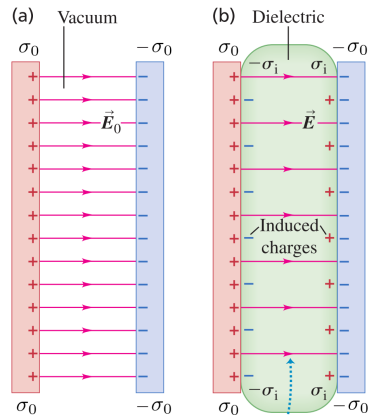


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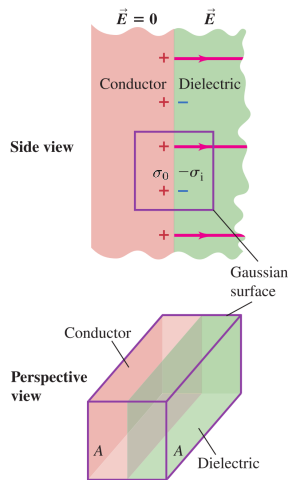
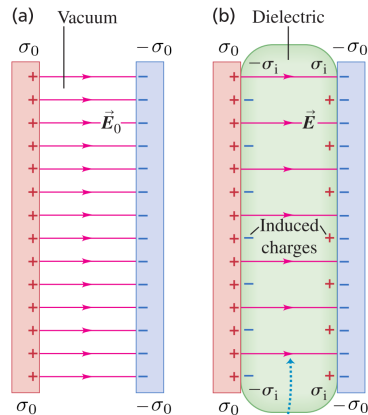


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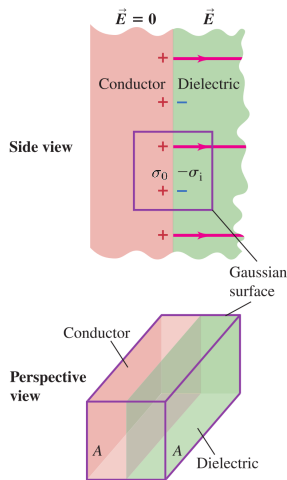
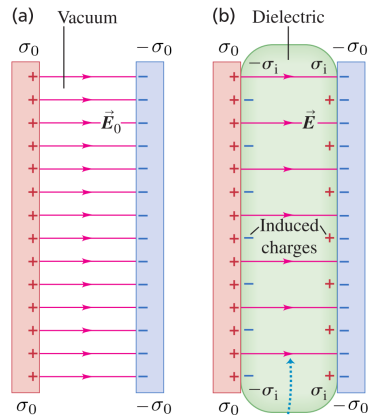


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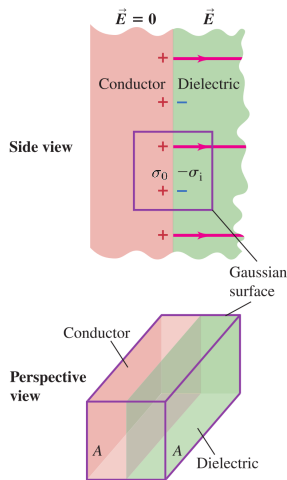
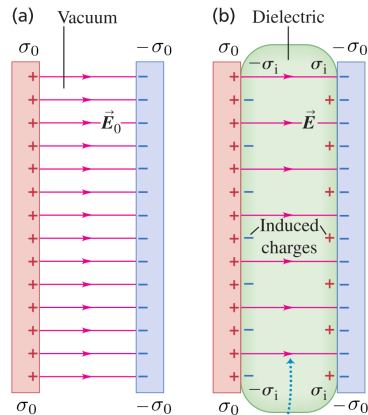


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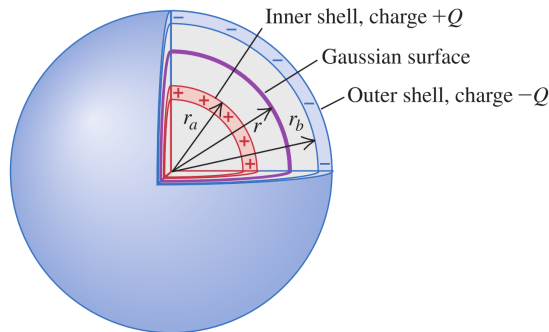
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Exercise I

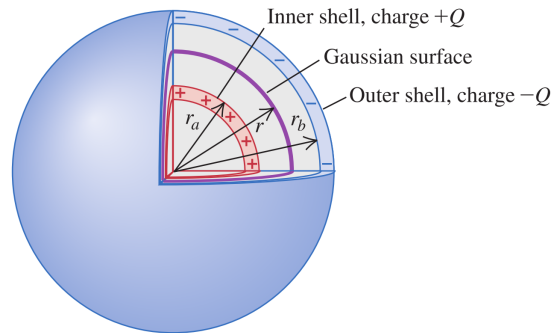
<https://epoll.srv.ualberta.ca> – Code: MBD

Use Gauss's law to find the capacitance of the spherical capacitor in the figure if the volume between the shells is filled with an insulating oil with dielectric constant K .



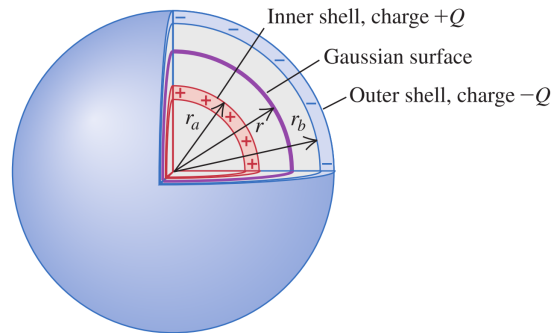
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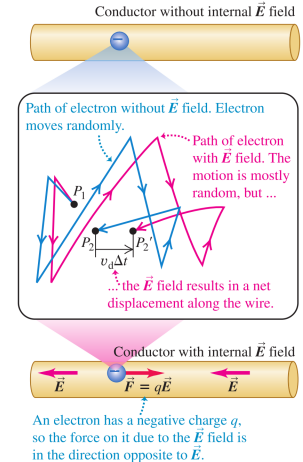
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Drift Velocity

In the absence of electric field inside a conductor

Figure 25.1 If there is no electric field inside a conductor, an electron moves randomly from point P_1 to point P_2 in a time Δt . If an electric field $\vec{E}$ is present, the electric force $\vec{F} = q\vec{E}$ imposes a small drift (greatly exaggerated here) that takes the electron to point P_2' , a distance $v_d \Delta t$ from P_2 in the direction of the force.

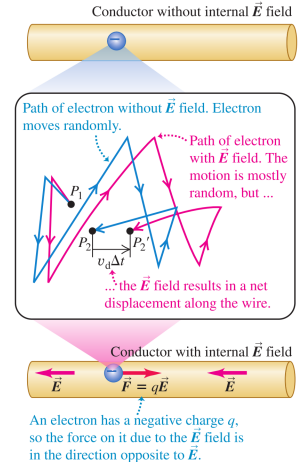


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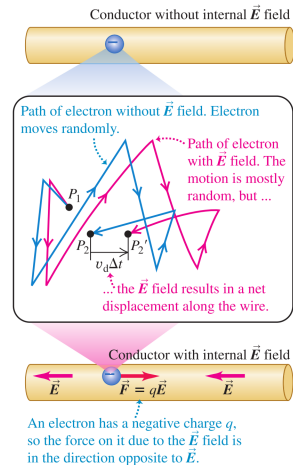


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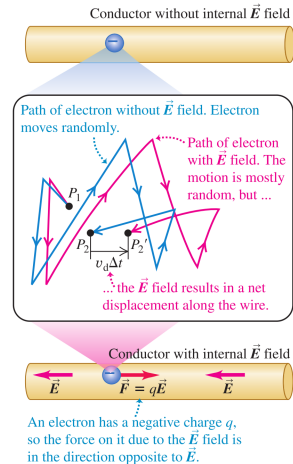


Drift Velocity

In the absence of electric field inside a conductor

- No external force acting on charges
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- But no net displacement, or flow of free charges (electrons)

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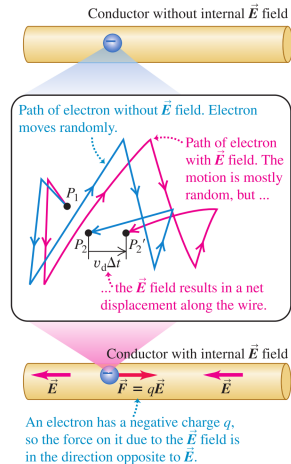
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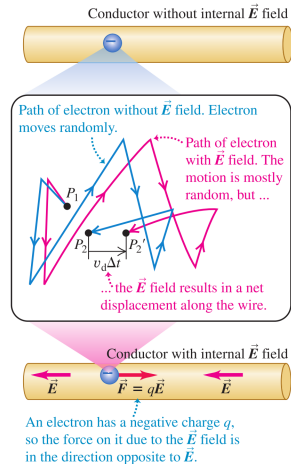
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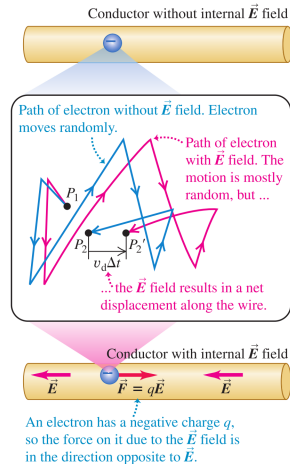
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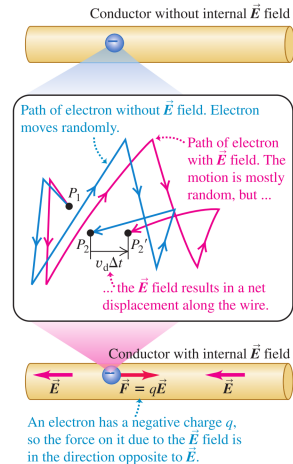
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- But there will be a net displacement, or slow flow of free charges (electrons)

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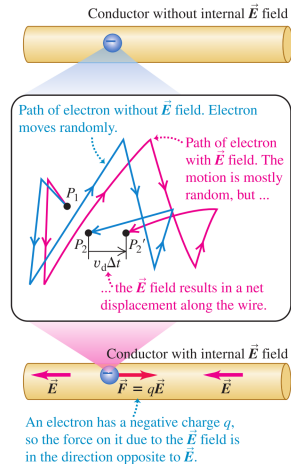
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- No external force acting on charges
- Free charges (electrons) move around randomly and collide with other charges
- But no net displacement, or flow of free charges (electrons)

If we create an $\mathbf{E}$ inside the conductor (yes we can do that! we will see later) $\Rightarrow$ **Electrodynamics, not electrostatics**

- A net external force $\mathbf{F} = q\mathbf{E}$ acting on charges
- Free charges (electrons) still collide with other charges
- But there will be a net displacement, or slow flow of free charges (electrons)
- This motion is described in terms of the drift velocity $\mathbf{v}_d$

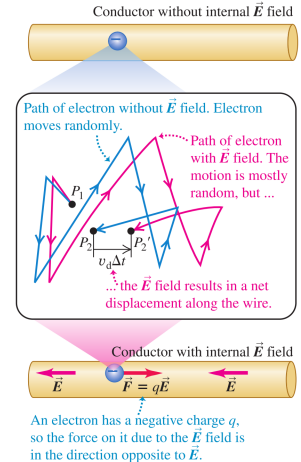
Figure 25.1 If there is no electric field inside a conductor, an electron moves randomly from point P_1 to point P_2 in a time Δt . If an electric field $\vec{E}$ is present, the electric force $\vec{F} = q\vec{E}$ imposes a small drift (greatly exaggerated here) that takes the electron to point P_2' , a distance $v_d \Delta t$ from P_2 in the direction of the force.



Electric Current

This flow of free charges (electrons) with $\mathbf{v}_d$ is called **electric current** !

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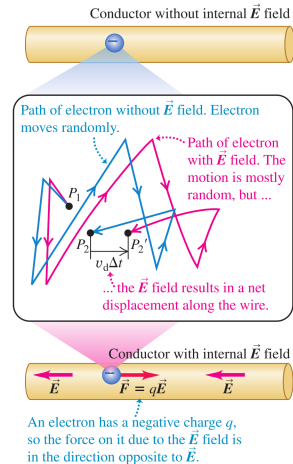


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- Random motion of electrons is very fast: $v = 10^6$ m/s

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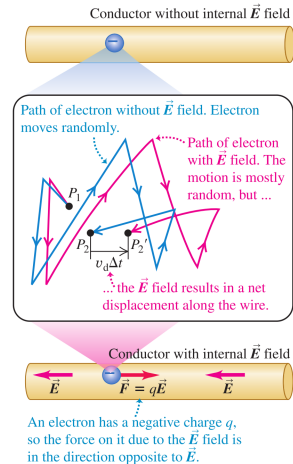


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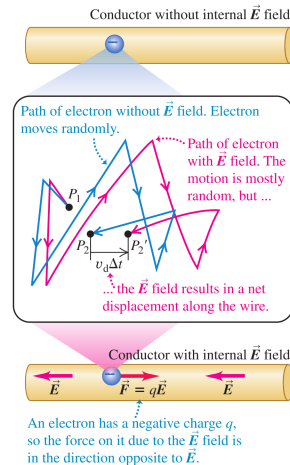
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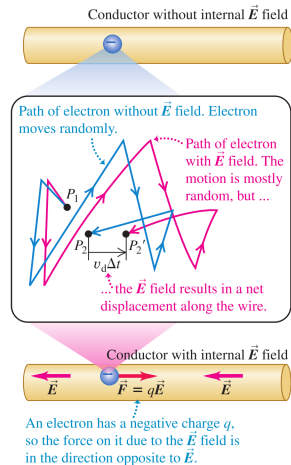
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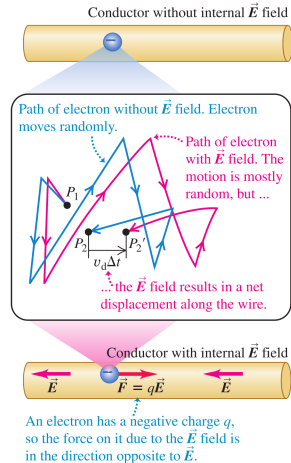
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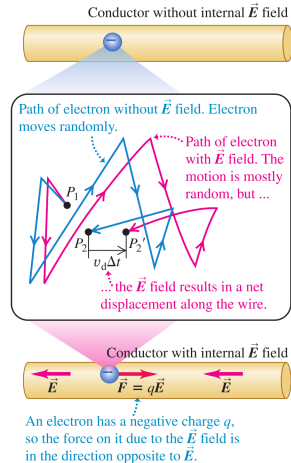
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- Electric field is established in the conductor with the speed of light (almost immediately)
- All electrons start moving with v_d , including the last one (closest to the lamp) $\Rightarrow$ reaches it almost immediately
- Like army soldiers in one line hearing the voice of the commander propagating with sound speed (like electric field propagating at speed of light) to start marching

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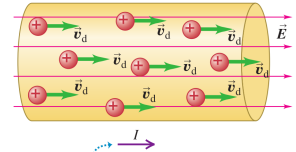


Electric Current

The **direction of electric current flow** is defined as the **direction of net flow of positive charges**

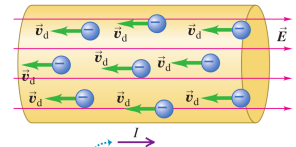
Figure 25.2 The same current is produced by (a) positive charges moving in the direction of the electric field $\vec{E}$ or (b) the same number of negative charges moving at the same speed in the direction opposite to $\vec{E}$.

(a)



A **conventional current** is treated as a flow of positive charges, regardless of whether the free charges in the conductor are positive, negative, or both.

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In a metallic conductor, the moving charges are electrons — but the **current** still points in the direction positive charges would flow.

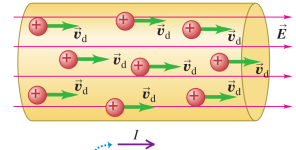
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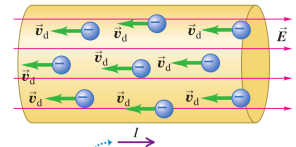
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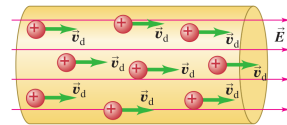
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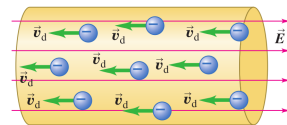
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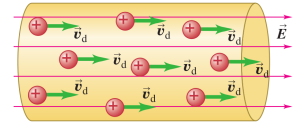
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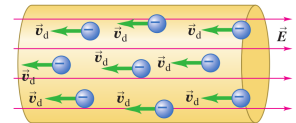
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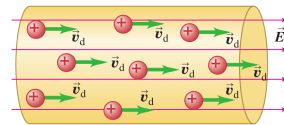
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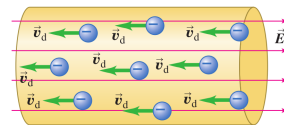
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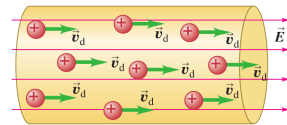
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- If positive charges are moving with $\mathbf{v}_d$: is the same as $\mathbf{v}_d$ (top picture)
- If negative charges are moving with $\mathbf{v}_d$: is opposite to the motion, i.e., $-\mathbf{v}_d$ (bottom picture)

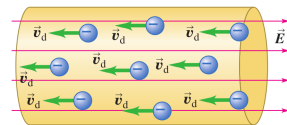
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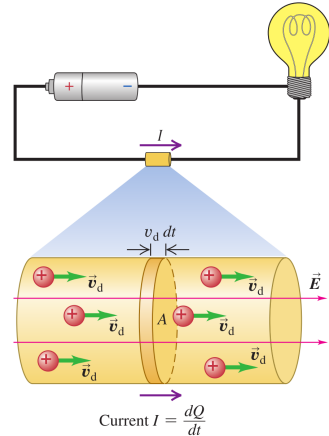
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Current I is the net charge flowing per unit cross-sectional area per unit time

$$I = \frac{dQ}{dt}$$

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).



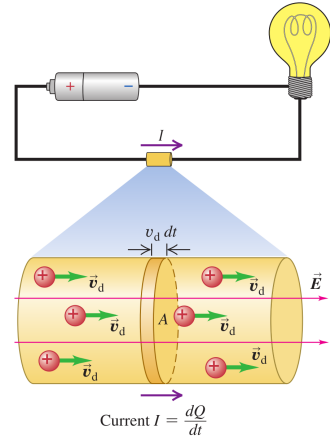
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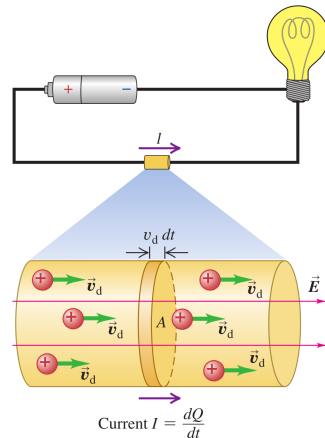
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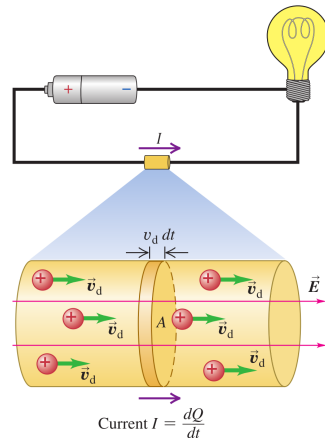
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Units of current is **ampere** A, which is also C/s

$$[I] = \frac{[dQ]}{[dt]} = \frac{\text{C}}{\text{s}} = \text{A}$$

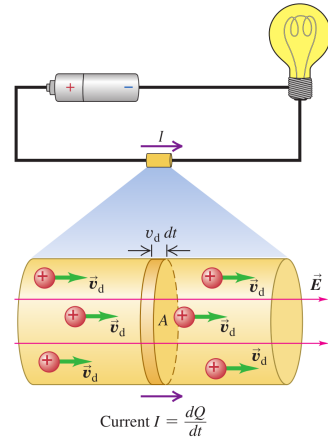
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Current Density

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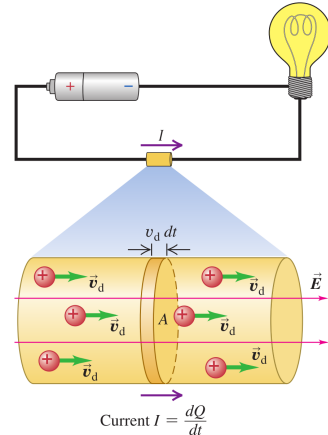
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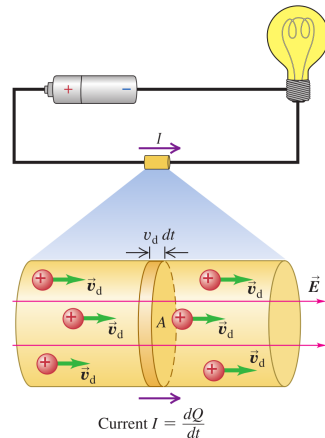
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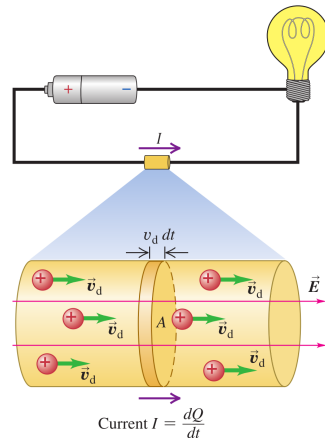
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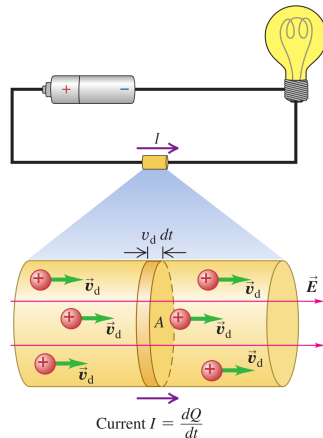
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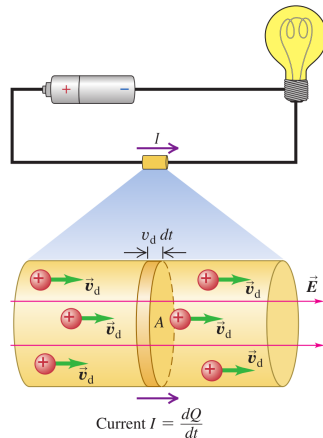
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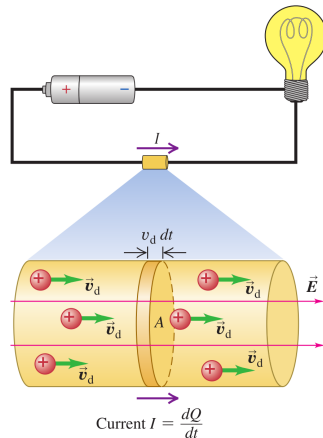
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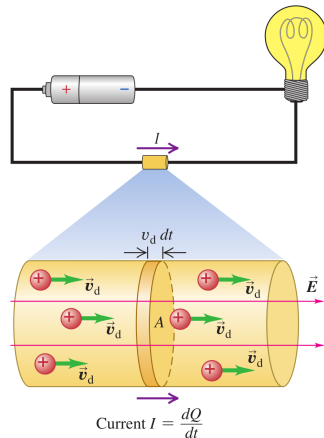
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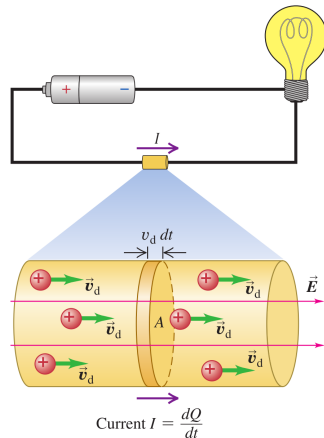
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Total charge that move from the left end to the right end in time dt is

$$dQ = Nq = nqAv_d dt$$

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).

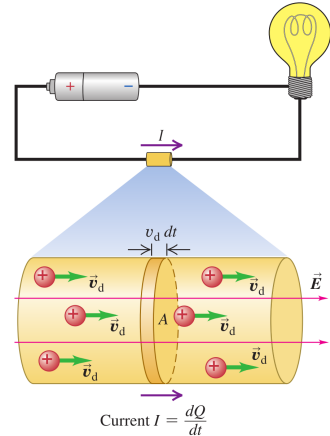


Current Density

So the total current I is

$$I = \frac{dQ}{dt}$$

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).

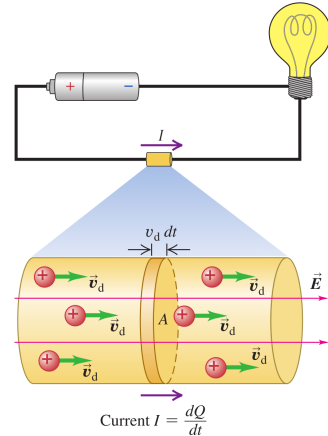


Current Density

So the total current I is

$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt}$$

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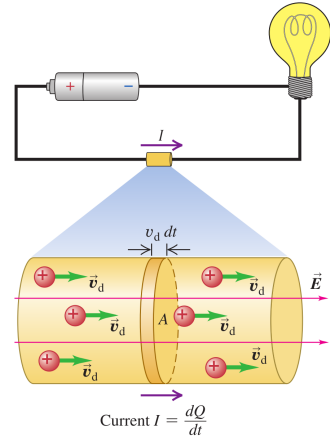


Current Density

So the total current I is

$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt} = nqAv_d$$

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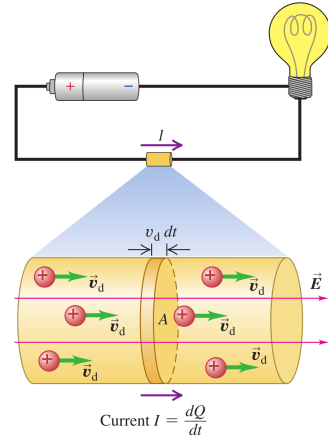
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The **current density vector $\mathbf{J}$** is defined as **current per unit cross sectional area in the direction of $\hat{\mathbf{v}}_d$** ,

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Current Density

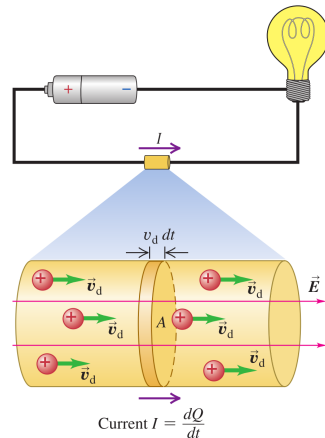
So the total current I is

$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt} = nqAv_d$$

The **current density vector $\mathbf{J}$** is defined as **current per unit cross sectional area in the direction of $\hat{\mathbf{v}}_d$** ,

$$\mathbf{J} = \frac{I}{A}$$

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).



Current Density

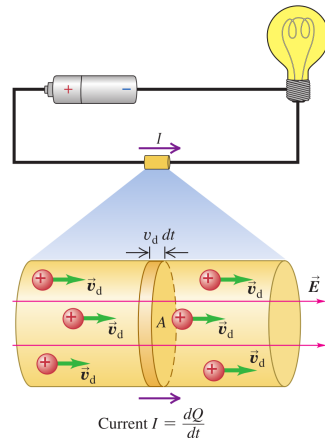
So the total current I is

$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt} = nqAv_d$$

The **current density vector $\mathbf{J}$** is defined as **current per unit cross sectional area in the direction of $\hat{\mathbf{v}}_d$** ,

$$\boxed{J = \frac{I}{A}} \Rightarrow \mathbf{J} = \frac{I}{A} \hat{\mathbf{v}}_d$$

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).



Current Density

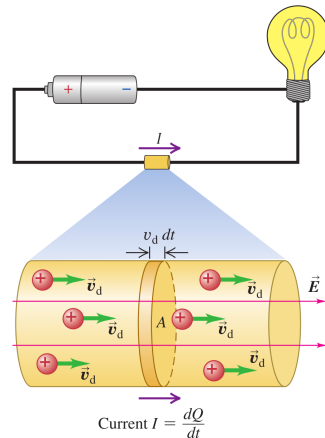
So the total current I is

$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt} = nqAv_d$$

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Current Density

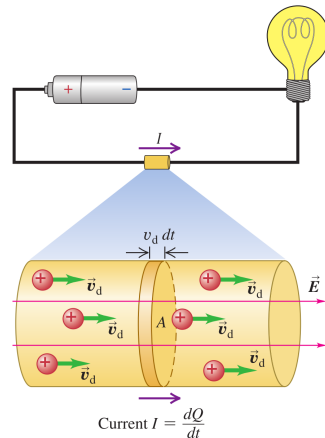
So the total current I is

$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt} = nqAv_d$$

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Current Density

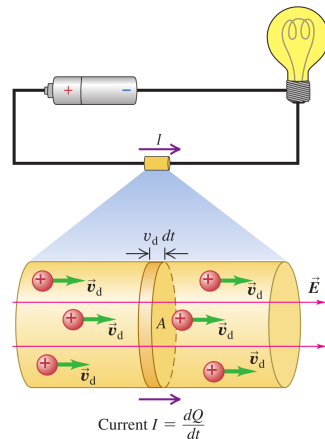
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$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt} = nqAv_d$$

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Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).



Current Density

So the total current I is

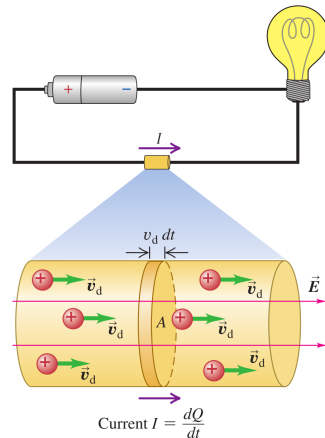
$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt} = nqAv_d$$

The **current density vector $\mathbf{J}$** is defined as **current per unit cross sectional area in the direction of $\hat{\mathbf{v}}_d$** ,

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If $q > 0$ then $\mathbf{J}$ is in the same direction as $\mathbf{v}_d$

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).



Current Density

So the total current I is

$$I = \frac{dQ}{dt} = nqAv_d \frac{dt}{dt} = nqAv_d$$

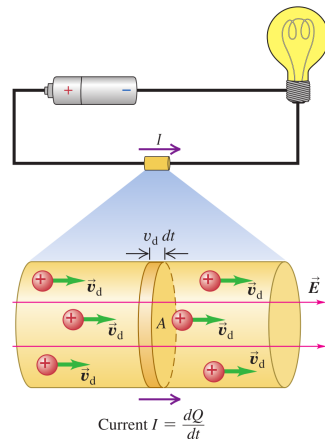
The **current density vector $\mathbf{J}$** is defined as **current per unit cross sectional area in the direction of $\hat{\mathbf{v}}_d$** ,

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If $q > 0$ then $\mathbf{J}$ is in the same direction as $\mathbf{v}_d$

If $q < 0$ then $\mathbf{J}$ is in the opposite direction of $\mathbf{v}_d$

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).

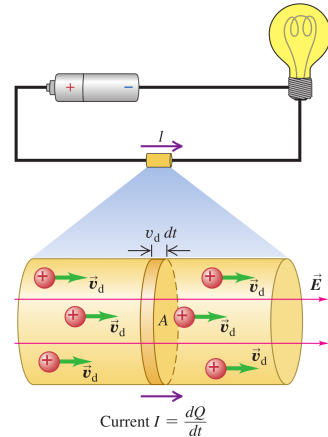


Current Density

Units of $\mathbf{J}$ is A/m^2

$$[\mathbf{J}] = \frac{[I]}{[A]} = \frac{\text{A}}{\text{m}^2}$$

Figure 25.3 The current I is the time rate of charge transfer through the cross-sectional area A . The random component of each moving charged particle's motion averages to zero, and the current is in the same direction as $\vec{E}$ whether the moving charges are positive (as shown here) or negative (see Fig. 25.2b).



Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD

A copper wire of diameter 0.98 mm carries a constant current of 0.26 A to a 60 W bulb. The free-electron number density in the wire is $8.5 \times 10^{28} \text{ m}^{-3}$. Find

(a) the current density magnitude

(b) the drift speed.

Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD